

The Oracle and the Kernel: A Falsifiable, Substrate-Bounded Theory of Domain Super-Intelligence

The Deterministic Oracle as Grounding Anchor, the Minimum Grounding Set as the One-Percent Kernel, and an Information-Theoretic Test that Decides Whether a Kernel Can Exist at All

Chris Royse

chrisroyseai@gmail.com

4 June 2026

Abstract. I define *operational super-intelligence in a domain* as the existence of a deterministic predictor that agrees with the domain’s ground-truth *oracle* at a calibrated threshold, stably, with Goodhart defenses and online mistake-closure—at which point human review of machine-generated artifacts in that domain becomes statistically unjustified. This is a falsifiable, oracle-bounded property of a specific predictor against a specific domain, not a claim about general cognition. The paper makes three moves. First, it places the *oracle* at the center: the oracle is the grounding anchor—the only point where the system touches non-linguistic reality—and its self-consistency is a hard ceiling on achievable correlation, decomposing into two independently violable axes, flakiness and validity (Aleithan et al., 2024; OpenAI, 2026; Wang, Pradel, & Liu, 2025). Second, it imports the symbol-grounding result from cognitive science—Harnad’s dictionary-go-round (1990) and the Vincent-Lamarre *minimum grounding set* of $\approx 1\%$ from a grounding kernel of $\approx 10\%$ (2016)—and shows a domain predictor is the same object: a web of associations anchored at a small grounding kernel, with the oracle as the anchoring channel and the kernel as the minimal set of artifacts sent through it. Third, the load-bearing contribution: a grounding kernel can exist only if the observation panel carries at least the oracle’s worth of information about the outcome, measured as mutual information $I(\text{panel}; \text{oracle})$. By the data-processing inequality (Cover & Thomas, 2006) this upper-bounds *every* predictor, kernel search, and calibrator reading the panel, so it falsifies an architecture before any model is trained. I instantiate the theory on ME-JEPA-Code, a binary Pass/Fail predictor for Python changes against a Docker test oracle (SWE-bench Lite, 300×8). The result is a clean negative the theory predicts and explains: the anchor is noiseless on the flakiness axis (*oracle_self_consistency* = 1.0), yet the nine-sensor panel carries only $I(\text{panel}; \text{oracle}) \approx 0.46$ of the ≈ 1 bit the verdict requires. A per-sensor decomposition localizes the deficit—the panel measures what code *looks like*, not what it *does*; no sensor is load-bearing; signal survives only in the single purely-syntactic class. The fix is not more training but a panel of outcome/causal and execution-derived sensors selected by *measured* bit-contribution. The general claim: domain super-intelligence reduces to a clean oracle, a panel sufficient to carry its bits, and the minimal grounded kernel from which the rest follows by association—each separately measurable, each separately falsifiable.

Keywords: symbol grounding; minimum grounding set; mutual information; data-processing inequality; oracle; calibrated verification; domain super-intelligence; JEPA.

1 Introduction

Machine generation of artifacts—code, proofs, images, diagnoses—has outrun human verification. A coding agent emits edits faster than a reviewer can read them, and the asymmetry widens with model capability. The binding constraint on machine throughput in any domain with a quality bar is therefore not generation but *verification*.

Verification can in principle be automated whenever the domain possesses a *deterministic oracle*: a machine-runnable function returning ground truth for a candidate artifact. Code has one (run the tests in a container); theorem proving has a proof checker; drug binding has a wet-lab assay; chess has engine search. The question this paper answers is: under what conditions can a learned predictor stand in for that oracle

so reliably that human review of machine-generated artifacts becomes statistically unjustified? A domain in which such a predictor exists is one in which *operational super-intelligence* has been reached, and I make the term falsifiable.

1.1 Thesis

Three linked claims, instantiated on a live system.

(1) **The oracle is the anchor, and its self-consistency is a ceiling.** A predictor can never agree with the oracle more often than the oracle agrees with itself. Oracle quality decomposes into two independently violable axes: *flakiness* (does the oracle reproduce its verdict?) and *validity* (does the verdict track the property of interest?), both documented in

the recent SWE-bench literature (Aleithan et al., 2024; OpenAI, 2026; Wang et al., 2025).

(2) A domain predictor is a grounded association engine. This is the symbol-grounding result transplanted. The dictionary-go-round of Harnad (1990) shows meaning defined only by reference to other meanings is circular; Vincent-Lamarre et al. (2016) make it quantitative—a definitional graph reduces to a grounding kernel of $\approx 10\%$ and a minimum grounding set (a minimum feedback vertex set) of $\approx 1\%$ from which all else is recoverable by association. The predictor is the same object, anchored at a small kernel of oracle-labeled artifacts. The *Vector Grounding Problem* of Coelho Mollo and Millière (2023) extends this from symbols to the vectors a panel computes over, supplying the criterion I adopt: a representation is referentially grounded when it stands in causal-informational relation to a worldly outcome *and* was selected for carrying that information.

(3) A kernel exists only if the panel carries the oracle’s bits. The mutual information $I(\text{panel}; \text{oracle})$ upper-bounds—by the data-processing inequality (Cover & Thomas, 2006)—what *any* predictor reading the panel can know about the outcome. If the panel shares fewer than the oracle’s worth of bits with the verdict, no predictor, kernel search, or calibrator can close the gap; the deficit is *upstream*, in the sensors. This converts “is this architecture viable?” into a cheap measurement that runs before any model is trained.

1.2 Contributions

- A falsifiable, oracle-bounded definition of domain super-intelligence (§3), as one predicate over six measurable tiers, with the correlation target explicitly capped by oracle self-consistency.
- A theory of the oracle as grounding anchor (§2), separating flakiness from validity and showing each bounds the predictor differently.
- The grounding-kernel reframe (§4): the dictionary $\approx 1\%$ minimum grounding set generalized to any domain, with the oracle as anchor and the kernel as the minimal set of artifacts to ground.
- The substrate-sufficiency test (§5): $I(\text{panel}; \text{oracle}) \geq \tau_{\text{MI}}$ as a method-independent existence test for a kernel, with the data-processing/Fano argument for why better modeling cannot evade it.
- An empirical instantiation with a clean negative result (§6–§7): on ME-JEPA-Code the oracle is noiseless on flakiness yet the panel carries ≈ 0.46 of ≈ 1 bit, and a per-sensor decomposition shows it measures form, not behavior.

- A method, not a guess, for fixing it (§6): assemble the panel from candidate sensors selected by measured marginal bit-contribution.

1.3 What is and is not claimed

Super-intelligence here is operational and per-domain: a predictor that replaces a human verification function at higher accuracy and machine throughput, against a domain with a deterministic oracle. It is not a claim about general intelligence, sentience, or any property lacking a deterministic oracle. The claim is falsifiable precisely because it is narrow, and the central empirical result of the instantiation is a *falsification*—the current panel provably cannot reach the gate—which is the kind of result the framework is built to produce early and cheaply. Association is circular unless grounded, so the entire edifice rests on the oracle being a faithful anchor; where the oracle is invalid or contaminated, apparent panel bits are bits about the wrong thing (§11).

2 The Oracle: The Grounding Anchor

2.1 Definition

For a domain D , an *oracle* is a deterministic, machine-runnable function $\text{Oracle}_D : \text{Artifact}_D \rightarrow \{\text{Pass}, \text{Fail}\}$ that returns ground truth without human judgment in the loop. The oracle is the single point at which the system touches *non-linguistic reality*; everything else is internal representation that acquires meaning only through its statistical relation to oracle outcomes. It is the *anchor* of §4: the channel through which a symbol (or vector) connects to something outside the symbol system. This is why the oracle, not the predictor, is the first object of study—a predictor is trained and judged against it, so if the oracle is the wrong instrument, every downstream number measures the wrong thing with spurious precision.

2.2 Self-consistency as a ceiling

The central structural fact: *you cannot agree with a measuring instrument more reliably than it agrees with itself.* Let *oracle_self_consistency* be the probability two independent evaluations on the same artifact agree. A predictor correlating at $\rho > \text{oracle_self_consistency}$ would have to predict the disputed artifacts more stably than the oracle’s own definition of truth—impossible. Hence

$$\tau_{\text{corr}}(D) \leq \text{oracle_self_consistency}(D). \quad (1)$$

Setting $\tau_{\text{corr}} = 0.95$ is coherent only if self-consistency ≥ 0.95 . Measuring it is therefore a cheap Tier-5 test that can invalidate the whole gate framing before any model work.

2.3 Two independent axes: flakiness and validity

“Oracle quality” is not one number.

Axis A—flakiness. Does the oracle reproduce its verdict on re-evaluation? Measured by replica voting (n runs per artifact, pairwise agreement). A flaky oracle caps τ_{corr} .

Axis B—validity. Does the verdict track the property of interest? An oracle can be perfectly reproducible yet systematically wrong: a weak test deterministically passes an incorrect artifact; an over-specified test deterministically fails a correct one. The SWE-bench audits quantify this on the most-used code oracle: [Aleithan et al. \(2024\)](#) report 32.67% solution leakage and 31.08% weak-test passes, with the same defects in Lite and Verified; [Wang et al. \(2025\)](#) show that running *all* repository tests flips 7.8% of plausible patches to incorrect; and [OpenAI \(2026\)](#) found, in an audit of a failed-task subset of SWE-bench Verified, 59.4% with flawed tests or problem statements. None are flakiness; all are validity failures.

A predictor that perfectly matches an invalid oracle has learned to reproduce its mistakes. High correlation against a contaminated or weak oracle is shared error, not super-intelligence—the honest boundary of the framework (§11): the claim is conditional on the oracle being a faithful anchor, and that condition is a measurement, not an assumption.

2.4 Contamination

Beyond flakiness and validity sits contamination: artifacts or gold solutions appearing in a sensor’s pre-training data ($> 94\%$ of SWE-bench issues predate common cutoffs ([Aleithan et al., 2024](#)); frontier models reproduce gold patches verbatim ([OpenAI, 2026](#))). Contamination inflates apparent panel sufficiency—a sensor may “carry bits” because it memorized the answer. Defenses: a per-sensor pretraining-overlap audit with a counterfactual-cutoff experiment, and cross-panel triangulation with disjoint encoders (§3).

3 The Universal Super-Intelligence Predicate

3.1 Objects and predicate

For domain D : the oracle Oracle_D (§2); a panel $\text{Panel}_D : \text{Artifact}_D \rightarrow \text{Array}\langle \text{Vec}_i \rangle$ packing an artifact into an array of per-sensor vectors, each in its own space (no flat concatenation; §6); a corpus Corpus_D over an (intervention $\times$ subdomain) matrix; and a deterministic learned Predictor_D . Then D admits superhuman binary verification iff there exists Predictor_D such that all of the following hold:

- T1.1** $\text{corr}(\text{Predictor}_D, \text{Oracle}_D) \geq \tau_{\text{corr}}$;
- T1.2** stable over k_{win} consecutive rolling windows;
- T1.3** $\text{corr}(\text{Predictor}_D^B, \text{Oracle}_D) \geq \tau_{\text{corr}}$ on a disjoint Panel B (Goodhart guard);
- T1.4** $|\text{corr}_A - \text{corr}_B| \leq \varepsilon_{\text{drift}}$;
- T2** the per-cell quality contract holds for every cell;
- T3.1** $\text{mistake_repeat_rate} \leq \delta_{\text{rep}}$;
- T3.2** $\text{unrelated_control_delta} \leq \varepsilon_{\text{unrel}}$;
- T4** every doctrine invariant equals its required value exactly;
- T5.1** $I(\text{Panel}_D; \text{Oracle}_D) \geq \tau_{\text{MI}}$ bits;
- T5.2** $\text{oracle_self_consistency} \geq \tau_{\text{sc}}$;
- T5.3** the inter-sensor distinctness gate passes for all slots;
- T6** any new/replaced sensor passes the novelty and utility gates.

When this is TRUE, human review of machine-generated artifacts in D is statistically unjustified. That is the operational definition of super-intelligence in D .

3.2 The six tiers

Tier 1 is the ship-gate: agreement at τ_{corr} , stable in time, surviving re-encoding through an independent Panel B without drift—the Goodhart guard ([Manheim & Garrabrant, 2018](#)). **Tier 2** requires the contract in *every* cell, not just aggregate (per-cell correlation, Brier ([Brier, 1950](#)), ECE ([Guo, Pleiss, Sun, & Weinberger, 2017](#)), class recalls with confidence-interval floors). **Tier 3** is online mistake-closure: a disagreement with reality becomes a training signal and does not recur, without degrading unrelated behavior. **Tier 4** are architectural invariants that must hold exactly (§3.4). **Tier 5** is substrate quality—panel sufficiency (§5), oracle reachability (§2), sensor non-aliasing—the cheapest, most decisive tests. **Tier 6** gates any sensor addition by novelty (not a near-duplicate) and utility (lifts per-cell correlation by a floor), the gate the assembly loop of §6 must satisfy.

3.3 Thresholds, and why these values

Thresholds are domain-set by the cost asymmetry between a false-Pass and a false-Fail, bounded above by the oracle (§2). For the Python instantiation: $\tau_{\text{corr}} = 0.95$, $k_{\text{win}} = 4$, $\varepsilon_{\text{drift}} = 0.05$, $\delta_{\text{rep}} = 0.05$, $\varepsilon_{\text{unrel}} = 0.02$, $\tau_{\text{MI}} = 0.95$ bits, $\tau_{\text{sc}} = 0.97$. **Revision is allowed only upward, and only on evidence:** if self-consistency is measured higher, τ_{corr} may rise; lowering it because the substrate cannot reach it is a definition change and a regression. The correct response to “we cannot reach 0.95” is to change *state* (better oracle, panel, corpus), never to move the goalpost.

3.4 The Tier-4 invariants (fences, not levers)

Eight invariants must hold exactly, because each, if violated, lets the system reach the numbers by cheating: (1) the binary contract; (2) slot-identity preservation; (3) no flat-vector collapse; (4) anti-compensation (a strong sensor may not silently cover a blind one); (5) frozen-target collapse-immunity (zero trainable target parameters); (6) no inner generator; (7) leakage policy (labels never enter as live inputs); (8) the grounded anchor (only the oracle is truth).

3.5 State is the lever; the predicate is the fence

The current corpus, panel, and harness are starting *state*, not constraints. Reaching the predicate may require corpus expansion, panel expansion, oracle refinement, or upward threshold revision—all legitimate. Only the Tier-4 invariants and the predicate are immovable. Five eligibility prerequisites precede the predicate: a deterministic oracle exists; it is reproducible and valid enough to set a coherent τ_{corr} ; artifacts are observable through sensors that can in principle carry outcome information; the matrix is enumerable; outcomes are feedable. §7 shows a domain can satisfy four of these and fail the third—the substrate prerequisite—which is exactly the negative result the framework surfaces early.

4 The Grounding Kernel

4.1 The dictionary-go-round and the $\approx 1\%$ result

Harnad posed the symbol grounding problem with an image (Harnad, 1990): learning Chinese as a first language from only a Chinese/Chinese dictionary is a merry-go-round from one meaningless symbol to another that never reaches meaning. The escape is to

ground a subset outside the symbol system. Vincent-Lamarre et al. made the structure exact (Vincent-Lamarre et al., 2016): modeling a dictionary as a directed graph (defining word $\rightarrow$ defined word), recursively removing words reachable by definition but defining nothing reduces it to a unique *Kernel* of $\approx 10\%$; the smallest set from which all the rest is definable—the *Minimum Feedback Vertex Set*, or *Minimal Grounding Set*—is $\approx 1\%$ of the dictionary ($\approx 15\%$ of the Kernel). Finding it is NP-hard (Karp, 1972). The interpretation: grounding one MinSet’s worth of words by sensorimotor contact suffices; the symbolic code defines the rest by recombination. Roughly 99% of meaning is carried by association; a $\approx 1\%$ grounded anchor carries the rest. The mental lexicon has the matching small-world, scale-free, differentiation-grown shape (Steyvers & Tenenbaum, 2005).

4.2 Anchor versus kernel

Two objects must stay separate. The *anchor* (grounding channel) is the operator that turns one item into a grounded fact by contact with reality—for the dictionary, sensorimotor experience; for a domain predictor, the *oracle*. The *grounding kernel* (MinSet) is the minimal *set of items* to send through the anchor so the rest follows by association. A system that sends *every* item through the anchor (oracle-everything, then correlate over the whole corpus) is not exploiting the kernel structure at all; it is brute-forcing the anchor across 100% of the space—exactly how a naive ship-gate behaves, and what the reframe replaces.

4.3 The kernel of an arbitrary domain

Transpose to domain D : *nodes* are candidate artifacts (or (artifact, verdict) pairs); the edge $X \leftarrow Y$ (“ X is definable from Y by association”) holds when, given the grounded verdict on Y , the panel predicts the verdict on X above a confidence threshold—computed in representation space (LeCun, 2022), concretely a constellation-membership test (§6). The kernel is the minimum feedback vertex set of this graph. Two consequences: (i) the kernel is the *hard boundary cases*, not the easy ones—obvious-Pass/obvious-Fail artifacts are the derivable 99%, while the kernel concentrates on the decision surface where only ground truth disambiguates (what an optimal active learner would query); (ii) there is no universal kernel across all knowledge—each domain has its own anchor and MinSet. What is universal is the *structure* (association graph + small grounded MinSet + anchor), not the kernel itself.

4.4 From symbols to vectors

A modern panel computes over vectors, so the relevant problem is the *Vector Grounding Problem* (Coelho Mollo & Millière, 2023): can internal vector states be *about* reality independently of human interpretation? Their teleosemantic answer, which I adopt as the sensor-admission criterion, is that a representation is referentially grounded when (1) it stands in causal-informational relation to a worldly outcome and (2) it has a selection history giving it the function of carrying that information. A sensor encoding mere surface similarity—selected to carry “what this looks like,” not “what outcome it produces”—is, for grounding purposes, ungrounded with respect to the outcome, however useful for retrieval. §7 shows this is exactly the measured deficit.

4.5 The reframe of the ship-gate

Combining §2 and §4 decomposes the naive gate into three separately-falsifiable questions, in dependency order: (1) *is the anchor clean?* (self-consistency, validity, contamination); (2) *can a kernel exist at all?*—does the panel carry the oracle’s bits, $I(\text{panel}; \text{oracle}) \geq \tau_{\text{MI}}$? if not, no kernel exists at this panel and the fix is upstream; (3) *what is the kernel, and how small?* The empirical contribution (§7) is that question 1 answers *yes* on flakiness and question 2 answers *no*.

5 Substrate Sufficiency: The Existence Test

5.1 The question and the ceiling

Let $Y = \text{Oracle}_D(x) \in \{\text{Pass}, \text{Fail}\}$ and $P = \text{Panel}_D(x)$. The outcome’s uncertainty is $H(Y)$ (≈ 1 bit when balanced); the bits the panel resolves are the mutual information $I(P; Y) = H(Y) - H(Y | P)$. The decisive property is method-independence: by the data-processing inequality (Cover & Thomas, 2006), for any function g ,

$$I(g(P); Y) \leq I(P; Y). \quad (2)$$

A predictor is such a g ; so is a constellation operator, a MinSet search, or a calibration map. *No processing downstream of the panel can manufacture information the panel does not contain.* Fano’s inequality (Cover & Thomas, 2006; Fano, 1961) turns this into an error floor: the minimum achievable error in predicting Y from P is bounded below by a function of $H(Y | P) = H(Y) - I(P; Y)$, regardless of the predictor. This is the information-bottleneck view (Tishby, Pereira, & Bialek, 1999): a panel that fails the test has discarded the relevant bits before the predictor sees them.

5.2 The predicate and decision rule

Define panel sufficiency as $I(\text{Panel}_D; \text{Oracle}_D) \geq \tau_{\text{MI}}$ (with $\tau_{\text{MI}} = 0.95$ bits). The rule:

Measured $I(P; Y)$	Verdict	Action
$\geq \tau_{\text{MI}}$	sufficient	gap (if any) is downstream
$[0.7, \tau_{\text{MI}})$	marginal	add 1–2 sensors; re-measure
< 0.7	insufficient	rebuild the panel

The test is gate-deciding and cheap relative to a training run: it can falsify an architecture before a single predictor is trained.

5.3 Estimation and decomposition

$I(P; Y)$ between a high-dimensional panel and a binary label is estimated with a stack read together, since each has a different bias/variance profile (Poole, Ozair, van den Oord, Alemi, & Tucker, 2019): **MINE** (Belghazi et al., 2018) (neural, Donsker–Varadhan, a lower bound); a **classifier-based lower bound** (a discriminator separating joint from product-of-marginals), validated by a synthetic self-test; **InfoNCE** (van den Oord, Li, & Vinyals, 2018) ($I \geq \log N - \mathcal{L}_{\text{NCE}}$); and the non-parametric **KSG** k -NN estimator (Kraskov, Stögbauer, & Grassberger, 2004). When these *agree* that $I(P; Y)$ is far below threshold the conclusion is robust, since lower bounds can only understate I . Two decompositions say *where* the deficit is: *per-cell* MI $I(P; Y | \text{cell})$ (does sufficiency fail uniformly or in specific cells?) and *drop-one-sensor* ablation $I(P; Y) - I(P \setminus \{s\}; Y)$ (is any sensor load-bearing?). Small drop-one deltas *with* low joint I indicate *absence* of signal, not redundancy.

5.4 Why sufficiency precedes kernel computation

A MinSet search is a particular $g(P)$, so it is bounded by $I(P; Y)$. If $I(P; Y) < \tau_{\text{MI}}$, the search cannot find a kernel that does not exist; it returns a degenerate “kernel” of essentially the whole corpus. The MI test is the cheaper, stronger existence test; the kernel computation finds the kernel once the MI test confirms one exists. The operational spine is therefore oracle $\rightarrow$ substrate $\rightarrow$ kernel $\rightarrow$ predictor/calibration.

5.5 Two gates: verdict-sufficiency and association-sufficiency

$I(P; Y) \geq \tau_{\text{MI}}$ is necessary but not the whole story. It guarantees the verdict is *recoverable*—the kernel *exists*—but not that the kernel is *small*. Two panels can both reach τ_{MI} yet differ sharply in geometry: in one,

the verdict is a smooth, local function of the panel (a candidate’s nearest neighbours share its label), so a tiny grounded set generalises and the MinSet approaches the dictionary’s $\approx 1\%$; in the other, the same bits are present but scrambled, the verdict is a wild function of position, and almost every point must be grounded (MinSet $\rightarrow$ the whole corpus). Kernel size is therefore set by a property beyond raw information: **association-sufficiency**—the degree to which same-verdict artifacts are co-located so the constellation-membership edge (§6) can actually derive one from another. It has a cheap proxy (k -nearest-neighbour verdict agreement in the panel, using same-sensor cosine per slot) and an exact measure (the MinSet fraction itself). Panel acceptance is thus *two* gates: **verdict-sufficiency** ($I(\text{panel}; \text{oracle}) \geq \tau_{\text{MI}}$, the oracle itself being validity-audited per §2) *and* **association-sufficiency** (high k -NN locality / small MinSet). A panel passing only the first can answer one artifact at a time but cannot ground a small kernel—which defeats the purpose.

6 The Panel and Its Sensors

6.1 An array of per-sensor vectors

The panel produces an *array* of per-sensor vectors, each in its own space. This is the JEPA stance (LeCun, 2022): heterogeneous encoders that need not share architecture or parameters, with prediction in representation space rather than by reconstruction. The license to treat N heterogeneous encoders as commensurable coordinates against one reality is the Platonic Representation Hypothesis (Huh, Cheung, Wang, & Isola, 2024): networks trained on different objectives and modalities converge to a shared statistical model of reality, with the converged kernel approaching the pointwise mutual information over the latent events that generate observations—the same PMI structure the grounding kernel and substrate test are built on.

6.2 Slot identity is sacred

Because each sensor’s vector lives in its own space, cross-sensor dimension-wise comparison is meaningless and forbidden (Tier-4). Permitted: same-sensor similarity; array-level structured operations; and scalar pairwise cross-sensor relationships (MI, correlation, agreement). The panel is *never* flattened into one vector for similarity, distance, OOD, calibration, or predictor input; flattening asserts a false commensurability between sensors and invites Goodhart. The measurements of §5–§7 are reported with `slot_identity_preserved=true` and `flat_vector_concat_used=false` as explicit invariants.

6.3 Constellation-as-definition

The association edge is realized as a constellation: the **Pass** class in a cell is defined by $\text{Def}(t) = (\{(c_m, \tau_m)\}_{m=1}^N, \{\rho_{jk}\}_{j < k})$ —per-sensor marginal regions (centroid c_m , threshold τ_m ; indexical) *and* the characteristic cross-sensor correlations ρ_{jk} (constitutive). Membership requires both marginals and the cross-sensor signature. This is the operator that lets one artifact’s verdict be derived from grounded neighbors—the edge the kernel computation runs on—and the “fast skill acquisition” reading of intelligence (Chollet, 2019) applied to a **Pass/Fail** region learned from $O(k)$ grounded examples.

6.4 Form versus behavior

The substrate test can reveal a panel referentially grounded in the wrong thing (Coelho Mollo & Millière, 2023): code encoders trained for *similarity/retrieval* encode what an artifact looks like, but a **Pass/Fail** oracle measures what it does when executed. The signature, visible in per-cell MI, is signal for *syntactic* interventions (a renamed variable changes tokens) and near-zero for *behavioral* ones (does it compile? was a call deleted? wrong file?). This splits the deficit: (1) **wrong objective**—fix with causal/outcome sensors trained on *edit* $\rightarrow$ *outcome*; and (2) **wrong modality**—static source under-determines runtime **Pass/Fail**, so the determining bits are absent from the input (the data-processing ceiling again), fixed with execution-derived sensors (traces, coverage, exception signatures, dependency/runtime/environment state).

6.5 Assembling a sufficient panel by measured bit-contribution

When the substrate test fails, select sensors by measured contribution to $I(\text{panel}; \text{oracle})$, not by guess: (1) acquire candidates across the missing axes; (2) run each over the corpus (slot identity preserved, no flat-concat); (3) measure standalone $I(\{s\}; Y)$, greedy marginal $\Delta I = I(P \cup \{s\}; Y) - I(P; Y)$, and per-cell ΔI , watching the behavioral cells at zero; (4) greedily assemble the panel maximizing $I(\text{panel}; \text{oracle})$ subject to the Tier-6 novelty and utility gates; (5) accept by re-running the substrate test, requiring $I(\text{panel}; \text{oracle}) \geq \tau_{\text{MI}}$ with behavioral-cell MI off zero. Two cautions make the search principled. First, mutual information is not additive across sensors, so selection must use the *marginal* ΔI in the context of the current panel, not standalone bits: redundant sensors carry the same bits (the failure mode of a panel of near-duplicate similarity encoders) and synergistic sensors carry little alone yet much in combination (so a low standalone score is not grounds to discard). Second, the cheapest gains are often *determin-*

istic structural features aimed at the cells reading zero bits—a parse/compile check for a compile-error class, a diff test for a deleted-call class, an import-closure test for a wrong-file class—added as isolated slots and gated per-cell so they cannot dilute the others. This is the teleosemantic grounding criterion (Coelho Mollo & Millière, 2023) made into a procedure: a sensor earns a slot when measured to carry outcome information and is kept because of that measurement. Only then is the kernel well-posed.

6.6 Predictor, calibration, online closure

Downstream of a sufficient panel, the predictor is a deterministic head over the panel and the constellation operator, emitting Pass/Fail with calibrated confidence (Brier (Brier, 1950), ECE and temperature scaling (Guo et al., 2017), conformal intervals (Angelopoulos & Bates, 2021) and an explicit abstain when out-of-distribution). Mistakes are fed back under protection against catastrophic forgetting (Kirkpatrick et al., 2017). None of this substitutes for substrate sufficiency: by the data-processing ceiling it is all bounded by $I(\text{panel}; \text{oracle})$. Calibration makes a sufficient panel’s predictor trustworthy; it cannot make an insufficient panel’s predictor correct.

7 Worked Instantiation: ME-JEPA-Code on Python

Every figure below is traceable to a source-of-truth artifact (§8); provenance is given as an FSV path under `/zfs/archive/contextgraph/...` or a tracked issue number.

7.1 Domain, oracle, corpus, panel

The domain is binary Pass/Fail prediction for a Python change; the oracle is a containerized SWE-bench harness (`run_evaluation`) (Jimenez et al., 2024); the corpus is SWE-bench Lite, 300 instances $\times$ 8 mutation categories = 2,400 candidates (categories: *known_good*, *subtle_flip*, *off_by_one*, *swap_variable*, *delete_test_call*, *wrong_file*, *over_engineer*, *compile_error*). The MI dataset has 2,400 rows, *feature_dim* = 19209, outcome entropy *target_bits* = 0.9978 (Pass 1267, Fail 1133). The panel has nine active sensors (`e1`, `e6`, `e7`, `e8`, `e9`, `e10`, `e12`, `e13`, `e14`), slot identity preserved, no flat-concat.

7.2 The anchor is clean on the flakiness axis

Replica voting (30 instances $\times$ 10 evaluations = 300 invocations) gives *oracle_self_consistency* = 1.0: every replica agreed; 300/300 reports; all exit codes 0. (Issue #760; FSV `inv-035-oracle-noise-floor-fsv/run-20260529T233417Z`.) This bounds *flakiness only* (§2): the oracle does not wobble, so it imposes no flakiness ceiling on τ_{corr} = 0.95. It does *not* establish *validity*; the weak-test/leakage hazards (Aleithan et al., 2024; OpenAI, 2026; Wang et al., 2025) are a separate, still-open axis (§11).

7.3 The substrate is insufficient

The panel-sufficiency probe ran over the 2,400-row dataset with the estimator stack (self-test: strong-signal control $\rightarrow$ 0.977 bits; noise $\rightarrow$ 0.0):

Estimator	$I(\text{panel}; \text{oracle})$	vs. 0.95
MINE (Donsker–Varadhan)	0.0960 bits	$\ll$
Classifier lower bound	0.1254 bits	$\ll$
Ross/Kraskov (k -NN)	0.4611 bits	$\ll$

Decision: *panel_sufficient*=false. Because these are lower-bound or conservative estimators, their agreement is robust: the panel carries at most $\approx$ 0.46 of the $\approx$ 1 bit required. (Issue #759; FSV `inv-032-panel-sufficiency-mi-fsv/run-20260529T234900Z/full-candidate-map`.) By the data-processing ceiling (§5), *no* predictor, kernel search, or calibrator reading this panel can reach 0.95. A useful grounding kernel does not exist at this panel; the deficit is upstream.

7.4 Where the bits are missing: form, not behavior

Per-cell MI: *swap_variable* = 0.2313 bits (the only non-trivial cell), *off_by_one* = 0.0207, and *compile_error*, *delete_test_call*, *wrong_file*, *over_engineer*, *subtle_flip* all $\approx$ 0. **Drop-one ablation (largest):** `e8` = 0.0309, `e6` = 0.0294, `e7` = 0.0214, `e14` = 0.0189 bits. (Same FSV run.) Two conclusions match §6: no sensor is load-bearing (largest contribution 0.031 bits, so *absence* of signal, not redundancy), and the panel sees form not behavior—signal only in the pure token-level cell, near-zero in the behavioral cells. This is the Vector Grounding Problem (Coelho Mollo & Millière, 2023) made empirical.

7.5 The predictor numbers, in their place

The predictor head achieves raw Pearson 0.4843 with the oracle, aggregate Brier 0.0924 and ECE 0.0240 (Brier, 1950; Guo et al., 2017); the nominal gap is $0.95 - 0.4843 = 0.4657$. (Ship-gate FSV `swebench-300x8-ship-gate-fsv/run-20260521T132308Z`.) Crucially this is not primarily a predictor-quality gap—training and calibration are bounded by $I(\text{panel}; \text{oracle}) \leq 0.46$ bits and cannot reach 0.95. A separate “rank-lift” diagnostic reporting 0.7942 is *not* the predictor’s oracle correlation; it is a category-local exemplar-similarity transform that lifts the apparent number without improving Brier or ECE, and is treated as a Goodhart artifact (Manheim & Garrabrant, 2018), not a gate pass.

7.6 Honest status

Gate	Status
Oracle flakiness (T5.2)	Pass (1.0)
Oracle validity	Open (weak-test/leakage)
Substrate sufficiency (T5.1)	Fail ($\leq 0.46 < 0.95$)
Kernel existence	No at this panel
Ship-gate corr. (T1.1)	Fail (0.4843)
T1.2–T1.4, T2, T3	not yet assessable

This is a clean negative result of the kind the framework is designed to produce early: the cheapest decisive test falsified the architecture before expensive model work, and the decomposition turned the falsification into a direction.

7.7 The forward plan

Per §6.5: acquire outcome/causal and execution-derived sensors, run them, measure standalone I and greedy ΔI with per-cell attention to the behavioral cells now at zero, assemble under the Tier-6 gates, and accept only when the substrate test returns ≥ 0.95 bits with behavioral-cell MI off zero. Only then is the kernel computation well-posed and the predictor able to move Tier-1. (Tracked: substrate reframe #842; kernel/MGS harness #843; panel sufficiency / in-slot sensor replacement #745; embedder candidate-screening loop #844; constellation-as-definition engine #841. The oracle-validity audit remains the most important open task.)

8 Verification Protocol

The framework’s claims are only as trustworthy as the discipline producing its numbers. The principle: *a return value is a claim; the source of truth is the*

verdict; read the verdict. Before a result is called done, four questions must have answers: what state changed; where it lives (a concrete source of truth); how it is read back independently; and the exact expected value. Three rules keep it honest: **no mocks** in gating tests (synthetic data only as an estimator self-test, never as evidence the system works); **fail closed** on unknowns with structured errors (the MI runner failed closed on degenerate all-Pass labels rather than fabricating a value); and **tests must fail when the feature is broken** (substrate sufficiency is designed to return false on an inadequate panel, and did). Each §7 figure is backed by a content-hashed artifact written to durable storage and independently re-opened for readback, so the number reported is the number in the hashed file. The default stance is adversarial: a result is guilty until proven innocent—which is why the 0.7942 rank-lift is flagged as Goodhart and oracle *validity* stays *open* despite flakiness measured at 1.0. This protocol is what makes the central claim—human review becomes statistically unjustified—a checkable statement rather than an act of faith.

9 Related Work

Symbol grounding. Harnad (1990), the quantitative kernel/MinSet of Vincent-Lamarre et al. (2016) (NP-hard per Karp, 1972), the small-world lexicon (Steyvers & Tenenbaum, 2005), and the vector-era update (Coelho Mollo & Millière, 2023). The contribution is to read a domain predictor as this object and make the kernel computable, with grounding turned from a diagnostic into a measurement and a selection procedure.

Information theory. The substrate test rests on the data-processing and Fano inequalities (Cover & Thomas, 2006; Fano, 1961) and the information bottleneck (Tishby et al., 1999), with estimators (Belghazi et al., 2018; Kraskov et al., 2004; Poole et al., 2019; van den Oord et al., 2018). The novelty is their use as a pre-training gate with per-cell/per-sensor diagnosis. The Platonic Representation Hypothesis (Huh et al., 2024) licenses reading a heterogeneous panel as coordinates against one PMI structure.

World models, calibration, abstention. The architecture is a JEPA (LeCun, 2022) extended with slot identity and the constellation operator; calibration and conformal methods (Angelopoulos & Bates, 2021; Brier, 1950; Guo et al., 2017) and continual learning (Kirkpatrick et al., 2017) are subordinated to the substrate test.

Benchmark integrity. The oracle-validity concern draws on (Aleithan et al., 2024; Jimenez et al., 2024; OpenAI, 2026; Wang et al., 2025); Goodhart’s variants (Manheim & Garrabrant, 2018) motivate the cross-

panel guard and the rejection of proxy metrics. The measure of intelligence of [Chollet \(2019\)](#) frames the constellation operator as few-shot region learning.

10 Generalization: From Verdict Oracles to World Models

The instantiation of §7 used a *binary verdict oracle*. The three conditions the framework reduces to—a clean oracle, a sufficient panel, a minimal grounded kernel (§4)—are not specific to binary verdicts. This section theorizes their extension to *predictive world models* in the sense of the Joint Embedding Predictive Architecture ([LeCun, 2022](#)).

Epistemic status. This section is more speculative than §1–§8: it is theory grounded in cited results and in the repository’s measured findings, not itself a measured result. It is included because the same three conditions, and the same substrate test, appear to govern world models, and because the failure mode measured for code (§7, “form not behavior”) is exactly the failure a world-model encoder exhibits when it captures appearance rather than dynamics. The genuinely uncertain ideas are given with pros and cons in §10.6.

10.1 The setting changes the oracle and the target

In the verdict setting the oracle returns one bit and $H(Y) \approx 1$ bit. In the world-model setting the “oracle” is reality’s response to intervention—act, the world evolves, observe—and the “verdict” is the high-dimensional future state. Hence the anchor is no longer clean (partially observable, stochastic, non-stationary); $\tau_{\text{MI}} = 0.95$ bits does not transfer and must be restated relative to how much of the future is predictable at all; and grounding requires *intervention*, since passive data under-determines causal structure.

10.2 The kernel becomes the predictive sufficient statistic

The dictionary’s Minimal Grounding Set (§4) is the minimal set for static *definability*. A world model needs the minimal set for *prediction of dynamics*, which is also rigorously defined. The **causal states** of a process—equivalence classes of histories inducing the same distribution over futures—form the *minimal sufficient statistic for prediction*, the provably simplest representation that loses no predictive power; the ϵ -machine over them is unique and optimal ([Shalizi & Crutchfield, 2001](#)). Its size is bounded by the **predictive information** $I_{\text{pred}} = I(\text{past}; \text{future})$, which is *sub-extensive* ([Bialek, Nemenman, & Tishby, 2001](#)): most of reality’s

bits are unpredictable detail, and only a small predictable core is learnable. That core is the world-model kernel. The dictionary’s “ $\approx 1\%$ ” figure is a property of dictionary-graph topology and does *not* transfer; what transfers is the structure (anchor + small grounded kernel + recovery by association) and the *existence* of a small predictable core.

10.3 The substrate test transfers directly

The sufficiency test (§5) generalizes from $I(\text{panel}; \text{oracle})$ to $I(\text{representation}; \text{future})$, bounded as before by the data-processing inequality ([Cover & Thomas, 2006](#)): no policy or planner reading the representation can know more about the future than the representation carries. The §7 finding—the code panel measures form, not behavior, so $I(\text{panel}; \text{oracle}) \approx 0.46$ —is the same failure a world-model encoder shows when it captures appearance rather than dynamics. JEPA’s bet ([LeCun, 2022](#)) (predict in representation space, discarding unpredictable detail) is, in this light, an attempt to learn the causal-state representation. The framework thus offers a concrete acceptance gate: measure $I(\text{representation}; \text{future})$ under intervention; if it is low while reconstruction loss is also low, the encoder learned form, not dynamics.

10.4 The grounded kernel has two layers

Layer A (prior kernel / start-up software): directly-grounded primitives, the analog of the dictionary’s grounded MinSet words—the core knowledge systems (objects, agents, number, space) ([Spelke & Kinzler, 2007](#)) and the developmental “start-up software” of intuitive physics and psychology ([Lake, Ullman, Tenenbaum, & Gershman, 2017](#)), seeded as inductive bias for sample efficiency. **Layer B (dynamical generators):** the minimal set whose closure generates predictable dynamics—symmetries and their conservation laws, a few interaction primitives at the agent’s scale, and the causal-intervention operators—i.e. the causal-state representation of the environment seeded by Layer A.

10.5 Two kernels: generative versus predictive

A natural objection is that the real kernel of the physical world is concrete—the building blocks themselves (subatomic particles, atoms, genes)—and that with enough microscopic detail one could predict everything (the reductionist / Laplace’s-demon hypothesis). The resolution is to separate two notions. The **genera-**

tive (ontological) kernel is the microscale rules plus the full microstate; its rules are genuinely tiny, and as a statement of what reality is *made of* reductionism is essentially correct. The **predictive (epistemic) kernel** is the smallest representation, at the agent’s scale and resources, retaining the predictable structure. The generative kernel is not a usable predictive kernel, and the predictive kernel is obtained by coarse-graining microscale detail *away*, not by accumulating it. Six independent obstacles block “predict the world from its microstate,” each sufficient alone: (i) reductionism does not imply constructionism—new organizing laws emerge at each scale (Anderson, 1972); (ii) simulating quantum matter is exponentially costly (Feynman, 1982); (iii) the microstate is unmeasurable in principle (uncertainty, no-cloning); (iv) chaos amplifies any micro-error, so micro-precision buys no macro-prediction; (v) computational irreducibility means there is generally no shortcut to running the system (Wolfram, 2002); (vi) the predictable structure is sub-extensive, so almost all microscopic bits are irrelevant (Bialek et al., 2001). The operation that legitimately turns microscale data into a kernel is coarse-graining—the renormalization group—and deep learning has been shown to implement a generalized RG-like coarse-graining (Mehta & Schwab, 2014), the mechanism by which a learner flows from fine input to the predictive kernel. The right scale is therefore not assumed but measured: the scale maximizing $I(\text{representation}; \text{future})$ per unit of representation cost. The honest concession is that this scale is domain-specific—for some domains (protein structure from sequence, material properties from atomic configurations with learned effective interactions) it sits near a physical-building-block level, partly vindicating the building-block instinct; for others (everyday manipulation, weather) it sits far above the microscale.

10.6 Pros and cons of the two conjectures

Conjecture 1—a world model is a kernel-finding problem (kernel = predictive sufficient statistic).

Pros	Cons
A small predictable core provably exists (predictive information is sub-extensive (Bialek et al., 2001); causal states are uniquely minimal (Shalizi & Crutchfield, 2001)).	The causal states are not given; reconstructing them in high dimensions is the learning problem itself.
Sample efficiency via a small kernel seeded by core-knowledge priors (Lake et al., 2017; Spelke & Kinzler, 2007).	The anchor is harder—partially observable, stochastic, requiring intervention to ground causal structure.
Measurability: the substrate test transfers as $I(\text{representation}; \text{future})$, a falsifiable gate.	The target is partly unpredictable (chaos, noise); the model is bounded by predictive information.
Transfer: a grounded dynamical kernel is a world model, not a task policy.	The kernel is scale- and goal-relative; the domain must be scoped.

Conjecture 2—the kernel could be a concrete physical-building-block scale (atoms / cells / genes).

Pros	Cons
Ontologically correct: the microscale plus its laws is a complete generator, and the laws are tiny—one basis in principle for all domains.	Reductionism is not constructionism (Anderson, 1972); levels cannot in practice be derived from below.
Empirically vindicated for specific domains (protein structure from sequence; material properties from atomic configurations).	Acquiring the microstate is forbidden in principle, and simulating it is exponentially costly (Feynman, 1982).
Avoids hand-designed abstractions; “learn from the bottom” is general.	Chaos and computational irreducibility (Wolfram, 2002) mean micro-precision does not buy macro-prediction; most micro-bits are irrelevant (Bialek et al., 2001).
Points to the correct mechanism—coarse-graining / RG (Mehta & Schwab, 2014).	Working systems already coarse-grain (learned effective maps); it is the wrong, intractable scale for most agent-scale tasks.

Synthesis. The generative kernel is real but unusable; the predictive kernel is what a world model needs; it is found by coarse-graining the microscale away and validated by the same $I(\cdot; \cdot)$ substrate test built for the verdict setting. The kernel’s scale is not assumed but measured—sometimes near the physical building blocks, usually above them. The framework’s contribution to the world-model question is therefore not a new architecture but a *measurement*: where, across scales,

the predictive information concentrates, and whether a candidate representation carries it.

11 Limitations and Scope

The claim is operational and per-domain, not general (§1). Its load-bearing limitation is that everything rests on the oracle being a faithful anchor: the Python instantiation measured oracle *flakiness* but has *not* bounded oracle *validity*, and the SWE-bench literature shows the validity hazard is large (Aleithan et al., 2024; OpenAI, 2026; Wang et al., 2025); a predictor that perfectly matched an invalid oracle would reproduce its mistakes, so a future Tier-1 pass must be read as conditional until validity is bounded. Contamination can masquerade as signal (Aleithan et al., 2024); in the current instantiation the panel is information-*starved*, so contamination is not inflating the present result, but it becomes first-order the moment $I(\text{panel}; \text{oracle})$ rises. The substrate estimate is a (robust) lower bound, decisive for *insufficiency* but requiring tighter estimation and a contamination audit for any future *sufficiency* claim; $\tau_{\text{MI}} = 0.95$ bits assumes a near-balanced outcome and must be restated relative to $H(Y)$ for skewed base rates. The $\approx 1\%$ kernel figure is specific to dictionary-graph topology (Vincent-Lamarre et al., 2016); the paper claims the *structure* is domain-general and the kernel *fraction* is measurable per domain, not that $\approx 1\%$ transfers. Permanently out of scope: domains without a deterministic oracle, ambiguous surfaces with no binary form against reality, generative substitution for the oracle, and threshold relaxation as a path to passing.

12 Conclusion

The bottleneck on machine output in any domain with a quality bar is verification, not generation, and it can be automated—to the point where human review becomes statistically unjustified—exactly when three separable, measurable conditions hold. The oracle is the anchor: the single contact with reality, with quality decomposing into flakiness and validity, neither an assumption. A domain predictor is a grounded association engine: a web of associations anchored at a small grounding kernel, the oracle the channel and the kernel the minimal set to send through it. And a kernel exists only if the panel carries the oracle’s bits—a mutual-information ceiling that bounds every predictor, kernel search, and calibrator, and so falsifies an architecture cheaply, before training. Instantiated on Python verification, the framework paid for itself with a clean negative: a noiseless anchor on flakiness, a panel carrying only ≈ 0.46 of ≈ 1 bit, and a per-sensor decomposition

showing the panel measures form, not behavior. The fix is prescribed without guessing—assemble a panel of outcome/causal and execution-derived sensors selected by measured bit-contribution, and only then compute the kernel. The general claim, restated: operational super-intelligence in a domain reduces to a clean oracle, a panel sufficient to carry its bits, and the minimal grounded kernel from which the rest follows by association—the same boundary Harnad drew thirty-five years ago, now with a number attached to it. The single most important open task is not a model improvement but a measurement: the oracle’s *validity*, so that a future gate pass means agreement with reality rather than with the oracle’s mistakes.

Competing interests

The author declares no competing financial interests. No vendor-funded benchmark or commercial-performance claim appears in this manuscript.

Reproducibility

Empirical figures trace to source-of-truth artifacts on the project’s durable host (§8), identified by FSV run path and tracked issue number. The mutual-information estimator stack includes a synthetic self-test (strong-signal control $\rightarrow 0.977$ bits; noise control $\rightarrow 0.0$) and fails closed on degenerate labels; all artifacts are content-hashed and independently re-opened for readback.

Companion documents

`docs/SuperIntelligence.md` states the universal predicate in full; `docs/paper_latex/main.tex` (“The Calculus of Association”) develops the frozen-instrument and constellation framing referenced in §6.

References

- Aleithan, R., Xue, H., Mohajer, M. M., Nnorom, E., Uddin, G., & Wang, S. (2024). SWE-bench+: Enhanced coding benchmark for LLMs. *arXiv preprint arXiv:2410.06992*. Retrieved from <https://arxiv.org/abs/2410.06992>
- Anderson, P. W. (1972). More is different. *Science*, 177(4047), 393–396. doi: 10.1126/science.177.4047.393
- Angelopoulos, A. N., & Bates, S. (2021). A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv preprint*

- arXiv:2107.07511. Retrieved from <https://arxiv.org/abs/2107.07511>
- Belghazi, M. I., Baratin, A., Rajeswar, S., Ozair, S., Bengio, Y., Courville, A., & Hjelm, R. D. (2018). Mutual information neural estimation. In *Proceedings of the 35th international conference on machine learning (icml)* (Vol. 80, pp. 531–540). Retrieved from <https://proceedings.mlr.press/v80/belghazi18a.html>
- Bialek, W., Nemenman, I., & Tishby, N. (2001). Predictability, complexity, and learning. *Neural Computation*, 13(11), 2409–2463. Retrieved from <https://arxiv.org/abs/physics/0007070> (arXiv:physics/0007070)
- Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1), 1–3. doi: 10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2
- Chollet, F. (2019). On the measure of intelligence. *arXiv preprint arXiv:1911.01547*. Retrieved from <https://arxiv.org/abs/1911.01547>
- Coelho Mollo, D., & Milli re, R. (2023). The vector grounding problem. *arXiv preprint arXiv:2304.01481*. Retrieved from <https://arxiv.org/abs/2304.01481>
- Cover, T. M., & Thomas, J. A. (2006). *Elements of information theory* (2nd ed.). Wiley-Interscience.
- Fano, R. M. (1961). *Transmission of information: A statistical theory of communications*. MIT Press.
- Feynman, R. P. (1982). Simulating physics with computers. *International Journal of Theoretical Physics*, 21(6–7), 467–488. doi: 10.1007/BF02650179
- Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. In *Proceedings of the 34th international conference on machine learning (icml)* (Vol. 70, pp. 1321–1330). Retrieved from <https://arxiv.org/abs/1706.04599>
- Harnad, S. (1990). The symbol grounding problem. *Physica D: Nonlinear Phenomena*, 42(1–3), 335–346. Retrieved from <https://www.southampton.ac.uk/~harnad/Papers/Harnad/harnad90.sgproblem.html> doi: 10.1016/0167-2789(90)90087-6
- Huh, M., Cheung, B., Wang, T., & Isola, P. (2024). The platonic representation hypothesis. In *Proceedings of the 41st international conference on machine learning (icml)* (Vol. 235, pp. 20617–20642). Retrieved from <https://arxiv.org/abs/2405.07987>
- Jimenez, C. E., Yang, J., Wettig, A., Yao, S., Pei, K., Press, O., & Narasimhan, K. (2024). SWE-bench: Can language models resolve real-world GitHub issues? In *International conference on learning representations (iclr)*. Retrieved from <https://arxiv.org/abs/2310.06770>
- Karp, R. M. (1972). Reducibility among combinatorial problems. In *Complexity of computer computations* (pp. 85–103). Plenum Press.
- Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., ... Hadsell, R. (2017). Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13), 3521–3526. doi: 10.1073/pnas.1611835114
- Kraskov, A., St gbauer, H., & Grassberger, P. (2004). Estimating mutual information. *Physical Review E*, 69(6), 066138. Retrieved from <https://journals.aps.org/pre/abstract/10.1103/PhysRevE.69.066138> doi: 10.1103/PhysRevE.69.066138
- Lake, B. M., Ullman, T. D., Tenenbaum, J. B., & Gershman, S. J. (2017). Building machines that learn and think like people. *Behavioral and Brain Sciences*, 40, e253. Retrieved from <https://arxiv.org/abs/1604.00289> (arXiv:1604.00289)
- LeCun, Y. (2022). *A path towards autonomous machine intelligence* (Tech. Rep. No. Version 0.9.2). OpenReview. Retrieved from <https://openreview.net/forum?id=BZ5a1r-kVsf> (OpenReview id BZ5a1r-kVsf)
- Manheim, D., & Garrabrant, S. (2018). Categorizing variants of goodhart’s law. *arXiv preprint arXiv:1803.04585*. Retrieved from <https://arxiv.org/abs/1803.04585>
- Mehta, P., & Schwab, D. J. (2014). An exact mapping between the variational renormalization group and deep learning. *arXiv preprint arXiv:1410.3831*. Retrieved from <https://arxiv.org/abs/1410.3831>
- OpenAI. (2026). *Why SWE-bench verified no longer measures frontier coding capabilities*. <https://openai.com/index/why-we-no-longer-evaluate-swe-bench-verified/>.
- Poole, B., Ozair, S., van den Oord, A., Alemi, A. A., & Tucker, G. (2019). On variational bounds of mutual information. In *Proceedings of the 36th international conference on machine learning (icml)* (Vol. 97). Retrieved from <https://arxiv.org/abs/1905.06922>
- Shalizi, C. R., & Crutchfield, J. P. (2001). Computational mechanics: Pattern and prediction, structure and simplicity. *Journal of Statistical Physics*, 104(3–4), 817–879. Retrieved from <https://arxiv.org/abs/cond-mat/9907176> (arXiv:cond-mat/9907176)
- Spelke, E. S., & Kinzler, K. D. (2007). Core knowledge. *Developmental Science*, 10(1), 89–96. doi: 10.1111/j.1467-7687.2007.00569.x
- Steyvers, M., & Tenenbaum, J. B. (2005). The large-scale structure of semantic networks: Statisti-

- cal analyses and a model of semantic growth. *Cognitive Science*, 29(1), 41–78. Retrieved from https://onlinelibrary.wiley.com/doi/10.1207/s15516709cog2901_3 doi: 10.1207/s15516709cog2901_3
- Tishby, N., Pereira, F. C., & Bialek, W. (1999). The information bottleneck method. In *Proceedings of the 37th allerton conference on communication, control and computing*. Retrieved from <https://arxiv.org/abs/physics/0004057>
- van den Oord, A., Li, Y., & Vinyals, O. (2018). Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*. Retrieved from <https://arxiv.org/abs/1807.03748>
- Vincent-Lamarre, P., Blondin Massé, A., Lopes, M., Lord, M., Marcotte, O., & Harnad, S. (2016). The latent structure of dictionaries. *Topics in Cognitive Science*, 8(3), 625–659. Retrieved from <https://onlinelibrary.wiley.com/doi/10.1111/tops.12211> doi: 10.1111/tops.12211
- Wang, Y., Pradel, M., & Liu, Z. (2025). Are “solved issues” in SWE-bench really solved correctly? an empirical study. *arXiv preprint arXiv:2503.15223*. Retrieved from <https://arxiv.org/abs/2503.15223>
- Wolfram, S. (2002). *A new kind of science*. Wolfram Media.